Solving the above equations gives

On , replace by . Now decide whether the wedge moves, using . If it does, then and require

**Else the wedge stays still; the block slides down it.**

**On , , using and the given data yields**

**On , use in in place of ; for , for . From and numerical data,**

**Thus wedge rests from to , but moves from to . The block's work on the wedge is**

is the wedge displacement;

**Find from vertical block motion from to , using and :**

**By kinematics,**

**: vertical part of just past . Obtain by energy conservation:**

Thus

Use in to find

**Take the positive sign and insert data:**

Use , recalling :

Use :